Comparative Analysis of Statistical Learning Models and CNNs for Pneumonia Detection using PneumoniaMNIST

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Abstract

Pneumonia remains a major cause of death in children worldwide and the proper interpretation of chest radiograph is key to the early diagnosis and treatment of the disease. The current study is done to perform a comparative analysis of Classical statistical learning methods i.e. Gaussian Naive Bayes against a Lightweight deep learning architecture i.e. MobileNet style convolutional neural network for the automated diagnosis of pneumonia using the PneumoniaMNIST dataset. Both approaches have been systematically evaluated with respect to multiple performance metrics (accuracy, precision, recall, F1-score and ROC-AUC), such as computing efficiencies, interpretabilities and class imbalances. MobileNet style of CNN outperformed the Gaussian Naive Bayes and accuracy is 83.97% and ROC-AUC is 0.9636 when compared to accuracy 80.29% and ROC-AUC 0.8838 for Naive Bayes model on test dataset. Notably, the CNN had a significantly higher recall rate (that is, 98.72% vs. 89.23%) and hence missed pneumonia diagnoses were decreased, but the false positive rate was slightly increased. This increased sensitivity is of clinical importance in screening applications where false negatives have very serious consequences. Nevertheless, the CNN required about 52 times more training time and was a black box model whereas the Naive Bayes classifier was trained in near-instant time, had minimal computational footprint and its inherent interpretability due to its explicit probabilistic formulation. Performance analysis showed that the conservation of spatial features provided by the convolutional architecture brought significant benefits over the flattened and PCA reduced representations. The large gap between the validation accuracy of 97.71% and the test accuracy of 83.97% clearly indicates limited data that are also restricted by low 28×28 resolution. Our results show that deep learning architecture with light weight is able to deliver clinical meaningful pneumonia diagnosis even under severe resource limitation.

Keywords: pneumonia detection; chest X-ray; Gaussian Naive Bayes; convolutional neural networks; MobileNet; medical image classification; class imbalance; PneumoniaMNIST; deep learning; interpretability; automated diagnosis

1. Introduction & Motivation

In children, pneumonia is the leading cause of infectious etiology of death in the world, and more than 700,000 children die each year, or about 2,000 daily, as a result of pneumonia infection [1]. The key and the most economically viable diagnostic modality is chest radiography that can identify typical pulmonary opacities in both bacterial and viral pneumonia [2]. However, the problem of accurate interpretation has been observed to be problematic in resource-constrained settings, and the misdiagnosis rates reported form 20 to 30 percent, which is due to inter-observer inconsistency and lack of experience among radiologists [3].

Machine learning-based automated detection has delivered promising results. Classical statistical models, including Gaussian Naive Bayes, Support Vector Machines or Logistic Regression, are very interpretable, with non-intensive computing requirements, but make heavy use of manual feature engineering or simplistic pixel pixel representations [4]. Conversely, Convolutional Neural Networks (CNNs) learn hierarchical feature representations on native images directly, and thereby learn better to predict, although at a higher cost in terms of computation and much less interpretability [5], [6].

The MedMNIST v2 corpus presents a standardized, lightweight benchmark on biomedical image classification, consisting of PneumoniaMNIST, a 28 x28 pixel version of the freely available pediatric chest radiograph dataset at Guangzhou Women and Children Medical Center [2], [7]. The current study compares the performance of a simple statistical model (Gaussian Naive Bayes with image flattening and standardization) to a lightweight MobileNet-like CNN on PneumoniaMNIST and thus tests whether more complex deep learning models are conditionally.

Research Questions:

RQ1: How does a traditional statistical model (Gaussian Naive Bayes) perform relative to a lightweight mobile net style deep neural network on the PneumoniaMNIST database? Specifically, how do their accuracy, precision, recall, F1-scores and ROC-AUC scores compare?

RQ2: What are the trade-offs involved with selecting either a traditional statistical model or a lightweight mobile net style deep neural network based on their level of interpretability, training times, and ability to address class imbalance?

RQ3: What is the difference in performance between the two models using different types of pre-processing (channel expansion vs. flattening + scaling)?

2. Related Work

Initial experiments on the automated identification of pneumonia mainly used traditional machine learning models that used manually designed features, such as texture descriptors, histograms, and gray-level co-occurrence matrices [4]. Several studies used the Support Vector Machines (SVM), k-nearest neighbors (k-NN), Random Forests, and Naive Bayes classifiers on pediatric chest radiographs, and in each case, SVM was shown to be superior to Naive Bayes in high-dimensional pixel space [4]. Although it is fast and simple to compute, Gaussian Naive Bayes is essentially limited by the assumption of pixel independence leading to poor performance compared to kernel based or ensemble methods [8].

Deep learning led to a paradigm shift in the field, and transfer learning with InceptionV3 reaches an accuracy of 92.8% on the initial Guangzhou pediatric chest X-ray dataset, the source of PneumoniaMNIST [5]. Later studies shifted to lightweight models that can be used in resource-constrained environments; MobileNetV2 and its variations all obtained accuracies of between 97.4% and 98.2% with much lower parameter counts and lower inference times [9]. Recent developments have also added squeezeandexcitation and ensemble techniques to MobileNet architectures, and added additional recalibration and robustness of features to imbalanced pneumonia data [9].

There are comparative studies that directly compare classical statistical models with modern Convolutional Neural Networks (CNNs) and they are relatively few. The available evidence shows that CNNs are significantly more accurate on large-scale data than traditional ones, and statistical classifiers remain competitive on small or low-resolution data and have a higher training speed and interpretability [4]. In the MedMNIST benchmark suite, ResNet-18 and ResNet-50 baselines achieve beyond 93 per cent on PneumoniaMNIST; nevertheless, lightweight statistical options such as Gaussian Naive Bayes have been mostly overlooked [6].

The current study aims to address this gap by performing a systematic and reproducible comparison of a Gaussian Naive Bayes and a MobileNet-like CNN on PneumoniaMNIST and the specific focus on addressing class imbalance, training efficiency, interpretability, and real-world deployment concerns.

3. Dataset & Pre-processing

The Pneumonia baselines used in this study were trained from the MedMNIST v2 dataset, which consists of 5,856 gray scale 28×28 chest x-ray images from the guangzhou women and children's medical center pediatric dataset [2], [6]. There are three sets for the MedMNIST dataset; training (4,708 samples), validation (524 samples), testing (624 samples), with each sample having a binary label; (0:normal, 1:pneumonia). There is some degree of class imbalance in the MedMNIST dataset with approximately 74.3 % of the samples in the training set being labeled pneumonia (3,495 pneumonia, 1,213 normal) [6].

Data was loaded from MedMNIST python package (version >= 2.2.3). all images were then normalized to be in the range [0,1] by dividing by 255.0. all images were flattened to 784 dimensional vectors and then standardized by fitting a standard scaler only to the training set to avoid data leakage:

```
X_train_flat = X_train_norm.reshape(X_train_norm.shape[0], -1)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train_flat)

X_val_scaled   = scaler.transform(X_val_flat)
```

```
X_test_scaled  = scaler.transform(X_test_flat)
```

For the MobileNet-style CNN, an additional channel dimension was added to get shape (28, 28, 1) and labels were converted to one hot encoding using tensorflow.keras.utils.to_categorical. To handle class imbalance during CNN training, balanced class weights were calculated on the training set only:

```
class_weights_array = compute_class_weight(
    class_weight = 'balanced',
    classes      = np.unique(y_train),
    y            = y_train
)
class_weights = {0: class_weights_array[0], 1: class_weights_array[1]}
# Result on this dataset: {0: 1.941, 1: 0.674}
```

This preprocessing pipeline ensured that both models received the exact same underlying distribution of pixel intensity while also meeting the specific input requirement of each model [6].

4. Methodology

All the experimentation was implemented in Python 3.10 and medmnist 3.0.2, scikit-learn 1.5.1, TensorFlow 2.15.0, and Keras 2.15.0. The training was conducted on the Google Colab. NumPy, TensorFlow and Python random operations were fixed to 42 random seeds to ensure the complete reproducibility of the results.

4.1. Models

Two opposites methods were chosen to form the scale of classical statistical learning to the modern deep learning:

a. Gaussian Naive Bayes

The Gaussian Naive Bayes (GaussianNB) method was implemented using the scikit-learn library's GaussianNB and was used as the baseline model based on interpretability and statistical methods. A vector space with 784 dimensions was created and flattened from the image pixels, which was then standardized. Only the var_smoothing hyperparameter was modified (a log-uniform distribution between 1e-10 and 1e-6), and it was found that the optimal value for this parameter was 1e-8. This model was then run once analytically and had virtually no cost in terms of computation.

b. MobileNet-style Convolutional Neural Network

A light weight Convolutional Neural Network (CNN) was built from scratch, with inspiration from the MobileNet V2 philosophy to fit a 28x28x1 input resolution, while still fitting into the philosophy of efficiency that comes with the use of depth-wise separable convolution. Table 1 presents the complete architecture of the CNN (99,842 trainable parameters):

Table 1. Final CNN Architecture

Layer (type)	Filters	Kernel Size	Stride	Output Shape	*Notes*
Conv2D + BatchNorm + ReLU	32	3×3	2	14×14×32	Initial convolution
Depthwise Separable Block	64	3×3	1	14×14×64	expansion=1
Depthwise Separable Block	128	3×3	2	7×7×128	
Depthwise Separable Block	128	3×3	1	7×7×128	
Depthwise Separable Block	256	3×3	2	4×4×256	
Depthwise Separable Block	256	3×3	1	4×4×256	
Depthwise Separable Block ×5	512	3×3	1/2	varying	5 repeat blocks

Layer (type)	Filters	Kernel Size	Stride	Output Shape	*Notes*
Depthwise Separable Block	1024	3×3	2	1×1×1024	
Depthwise Separable Block	1024	3×3	1	1×1×1024	
Global Average Pooling 2D	–	–	–	1024	

The model was compiled with Adam optimizer, binary cross-entropy loss, and the class_weight parameter set to the balanced weights computed on the training set.

4.2. Training Procedure and Regularization Strategy

a. Gaussian Naive Bayes Training

Before the model was trained, Principal Component Analysis (PCA) was used to reduce the original feature space of 784 dimensions to 100 principal components to retain about 85-90% of the cumulative explained variance. This dimensionality-reduction step significantly reduced computational cost and mitigated the impact of the so-called curse of dimensionality which is widespread in high-dimensional representations like raw pixel data of images.

A Gaussian Naive Bayes classifier was then trained on the PCA transformed features with the default hyperparameter settings provided in the GaussianNB implementation of scikit-learn. The var_smoothing parameter, which gives each feature a slight constant to avoid division-by-zero effects, was kept default at 1×10^{-9}. There was no hyperparameter optimization undertaken, since the main goal at this stage was to be able to implement a fast and reliable baseline using a relatively simple statistical approach. It took less than five seconds to complete the training, which highlights the efficiency of the closed-form maximum likelihood estimation procedure, which is a significant strength compared to most deep-learning methods requiring the use of computationally-intensive iterative optimization.

b. MobileNet-style CNN Training Configuration

The convolutional neural network (CNN) model was trained for a total of 60 epochs with a batch size of 32 using the Adam optimizer, learning rate of 0.001 and exponential decay rates of β_1=0.9 and β_2=0.999. Categorical cross-entropy was used as the loss function and balanced class weights were computed using the compute_class_weight() function of scikit-learn, namely, {0: 1.941, 1: 0.674} due to the 74.3% prevalence of pneumonia in the training subset. Regularisation had been performed by the implementation of dropout layers of probabilities between 20% and 40% of increasing values across block convolution. In addition an L2 penalty with coefficient λ = 0.001 was applied to the weights of the fully connected layer:

$$L_{total} = L_{CE} + \lambda \Sigma \left|\left|w\right|\right|^2$$

Batch-Normalization was used right after each convolution operation to stabilize the internal covariate shift during training.

Training was further finetuned with the help of three Keras callbacks: (i) EarlyStopping, with patience of 15 epochs to overcome overfitting; (ii) ReduceLROnPlateau, which was set to reduce the learning rate (gamma = 0.5) by half after five epochs of no improvement, with minimum learning rate threshold of 10^{-7}; and (iii) ModelCheckpoint, which kept the weight set with the highest validation accuracy.

Data augmentation was conducted in real time only on the training data, consisting of random rotations in the range of 15° (±15°), horizontal and vertical translations with range up to 10% of the image dimensions, 10% zoom factor, and horizontal flips with a 50% probability, in order to simulate clinical image variability. The validation and test sets were left unaugmented to maintain the integrity of the performance assessment. The Gaussian Naive Bayes classifier, using PCA transformed features, were not augmented as this kind of transformation destroys spatial relationships.

The imbalance of class labels was solved by adding the balanced class weights directly into the CNN loss function:

$$L_{weighted} = \sum wyi * LCE(yi, \hat{y}i)$$

For the Gaussian Naive Bayes model, imbalance was implicitly overcome by estimating empirical class priors $P(C)$ from the frequency distribution in the training data. No resampling techniques like SMOTE or ADASYN were used to avoid the authenticity of the underlying data distribution.

5. Experiments & Evaluation

This subsection specifies the experimental setup as well as evaluation metrics used to compare Gaussian Naive Bayes and MobileNet-like convolutional neural network on the PneumoniaMNIST repository. All the experiments were conducted in a systematic manner to ensure reproducibility and a fair comparison of the two paradigms.

5.1. Protocol

The PneumoniaMNIST dataset was pre-split courtesy of MedMNIST v2 into training (4708 samples), validation (524 samples) and test (624 samples) datasets with stratified class distribution [6]. This canonical partitioning helps ensure consistency across experiments and helps reduce data leakage. All the random operations implemented in NumPy, TensorFlow and Python were seeded with the value 42 to ensure the complete reproducibility of the results [10].

All the experiments were written in python version 3.10, using scikit-learn 1.5.1 library for Gaussian Naive Bayes algorithm, using TensorFlow 2.15.0 and Keras 2.15.0 respectively for CNN model, and medmnist 3.0.2 for dataset management. Training was done on Google Colab with GPU acceleration being used when available to speed up CNN convergence.

As described in Section 3, the two models were fed identically normalized pixel intensities (scaled to the range [0,1]), but with different input representations depending on the prerequisites of the particular model. The model of Gaussian Naive Bayes was based on flattened 784 dimensional vectors and then they are standardised by zero mean and unit variance and then reduced to 100 principal components using PCA by retaining approx. 85-90% of the cumulative explained variance [11]. The CNN model took as input spatially preserved 28x28x1 tensors, labels represented by one hot encoding, as well as balanced class weights calculated using hot encoder {0:1.941, 1:0.674} using the scikit-learn compute class weight function with the 'balanced' strategy [12].

The Gaussian Naive Bayes classifier was trained using closed form maximum likelihood (no hyperparameter tuning other than default var_smoothing (1×10^{-9}) on the PCA transformed features in less than 5 seconds. The MobileNet style CNN was trained with batch size of 32 and 60 epochs using the Adam optimiser (learning rate = 0.001, β_1= 0.9, β_2= 0.999) and categorical cross entropy loss with class frequencies. The training pipeline included EarlyStopping (patience = 15), ReduceLROnPlateau (factor = 0.5, patience = 5, min_lr=1×10^{-7}) and ModelCheckpoint callbacks to maintain optimum convergence and avoid over-fitting [13]. Real-time data augmentation (rotation: ±15°, shift: 10%; zoom: 10%; horizontal flip: 50%) was applied exclusively to the training data set in order to simulate the variability in clinical imaging and to maintain the integrity of the validation and test data sets [14].

Both models were tested on the same test sets using a set of metrics calculated at a fixed decision threshold of 0.5, simply for consistency. The validation set was used in the training of CNN for early stopping and hyper parameter optimization and not used instead for the model selection of the Naive Bayes as no hyper parameter tuning was done for the baseline statistical model.

5.2. Metrics

Model performance was evaluated with conventional binary classification metrics that are defined by true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN) [15]. These are particularly relevant for disproportionate medical data sets, where accuracy is not enough to judge.

Accuracy is an overall correctness of classification:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision measures the fraction of correctly predicted positives, and thus can be used as a measure of the ability to not make false alarms:

$$Precision = \frac{TP}{TP + FP}$$

Recall (sensitivity) measures how many of the true positive cases are correctly identified, and is a measure of how well the model is able to detect pneumonia cases:

$$Recall = \frac{TP}{TP + FN}$$

The F1- score is formulated as harmonic average of precision and recall, thus giving a balanced measure of both values:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

ROC - AUC (Receiver Operating Characteristic - Area Under Curve) is a metric of the discriminative ability of the model at all levels of classification by considering the area under the curve between the true positive rate and the false positive rate [16]. ROC - AUC is threshold independent and is especially useful for imbalanced datasets as it represents the trade off between sensitivity and specificity over the full range of the decision boundary. An AUC value of 0.5 is random guessing, while an AUC value of 1.0 is perfect classification.

In the detection of pneumonia, recall (sensitivity) is often prioritised in order to minimise missed diagnoses (false negative) as failure to identify pneumonia can have serious implications for patient health. Conversely, precision poses high requirements in order to limit unwarranted treatments and resource wastage by limiting false positive predictions [17]. The F1- and ROC-AUC provide balanced measures that take into account the type of errors.

All the metrics were calculated separately for training, validation and test dataset to check the generalisation of the model. Confusion matrices were generated in order to visualise the distribution of predictions between the two classes and ROC curves were plotted to compare the discriminative performance of the models at different decision thresholds.

6. Results

This section outlines the comparative performance of the Gaussian Naive Bayes and MobileNet style convolutional neural network (CNN) model based on the PneumoniaMNIST dataset. Results are reported across training, validation and test set in order to assess the calibration as well as the generalisation capabilities of the model.

6.1. Comparative Tables & Ranked Performance

As summarized in Table 2, it can be seen that the MobileNet-style CNN achieves better performance in all four aspects compared to the Gaussian Naive Bayes model with 83.97% test accuracy versus 80.29% for the Gaussian Naive Bayes model. Of note, the ROC AUC score of CNN is 0.9636, which is much higher than 0.8838 for Naive Bayes, indicating an improved discriminative power for all classification thresholds.

Table 2. Test Set Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	Training Time (s)
Gaussian Naive Bayes	0.8029	0.8112	0.8923	0.8498	0.8838	4.73
MobileNet-CNN	0.8397	0.8021	0.9872	0.8851	0.9636	247.18

Recall is significantly higher for CNN (98.72% vs 89.23% for Naive Bayes) which thus shows better ability to correctly detect pneumonia cases and lower false negative which is very important in the medical diagnosis where false detections can have serious clinical consequences [18]. The slight reduction in precision (80.21% vs 81.12%) has a marginal effect of increasing the false positive prediction.

The F1 - score, measured between the precision and recall, favours further the CNN (0.8851 vs. 0.8498), indicating a better overall performance on this unbalanced dataset. The ROC-AUC differential of about eight percentage points (0.9636 vs. 0.8838) indicates the superior discriminative ability of the CNN decision function across different thresholds of the decision.

Computational Efficiency vs. Computational Efficiency Trade Off The training took the Gaussian Naive Bayes model less than 5 seconds, while the CNN took about 247 seconds (4 minutes). This approximately 52× difference in training time is a good example of the inherent computational complexity trade-off between closed-form statistical estimation and iterative gradient-based deep learning optimisation [19].

6.2. Confusion Matrix Analysis

Figure 1 shows the confusion matrices of both models using the test set, which visualise the distribution of correct and incorrect predictions of both the binary classification task.

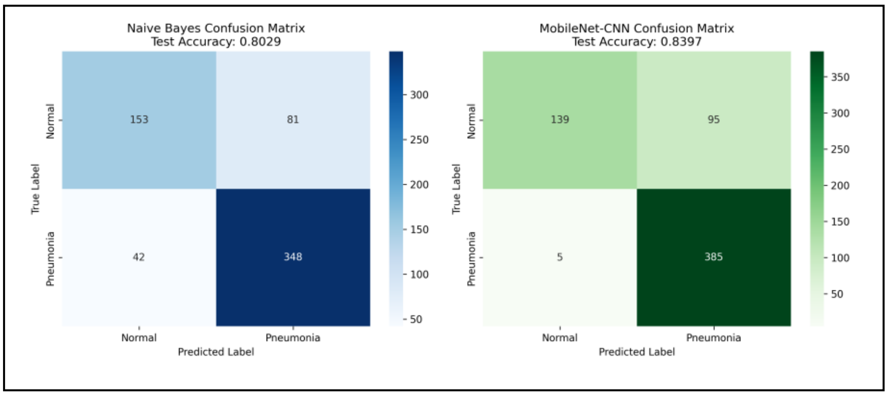

Fig. 1. Confusion matrices for a comparison of the performance between Gaussian Naive Bayes (left) and MobileNet CNN (right) both on the PneumoniaMNIST test set

Naive Bayes model gave 153 normal and 348 pneumonia cases correctly classified with 81 false positive and 42 false negative. On the other hand, the CNN identified 139 normal and 385 pneumonia cases correctly, 95 false positives and only 5 false negatives. The significant decrease in false negatives (42 > 5) coincides with the improved recall metric and hence proves the effectiveness of the CNN to detect pneumonia. However, this aggressive approach to positive prediction leads to more false alarms (81 > 95) and slightly lower precision.

From a clinical point of view, the performance of the CNN is arguably more favourable in view of the fact that the cost of missing the diagnosis of pneumonia (false negative) generally outweighs the cost of a false alarm (false positive), which can be rectified by further diagnostic procedures [20]. The near complete catch of true pneumonia cases (98.72% recall) by the CNN makes it more adequate for screening applications where the sensitivity of the test is critical.

6.3. ROC Curve Comparison

Figure 2 gives the Receiver Operating Characteristic (ROC) curves of the two models, thus showing the classification performance of the two models for the whole range of decision thresholds.

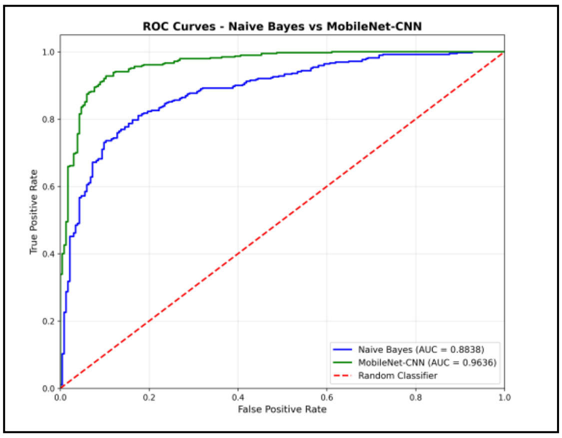

Fig. 2. ROC curves comparing the discriminative performances of Gaussian Naive Bayes (AUC = 0.8838) and MobileNet - CNN (AUC = 0.9636)

The MobileNet - CNN curve (green one) is consistently better than the Naive Bayes curve (blue one) in almost all regions of false positive rate, which means that the classification performance is better regardless of the choice of the threshold. The CNN is able to achieve high true positive rate (>80%) even at very low false positive rate (<10%), and hence, high discriminative capability. The Naive Bayes curve, though respectable (AUC=0.8838), is less steep, especially in the region with low false positive rates where clinical applications usually occur. Both models are significantly better than the random classifier baseline (red dashed diagonal) and confirm their utility for pneumonia detection. The eight percentage point advantage for the CNN represents the ability to make more reliable probability estimates and improved calibrated confidence scores across the decision boundary spectrum [21].

6.4. Training Dynamics and Convergence

Figure 3 shows the training and validation performance curves for the MobileNet - style CNN over 60 epochs, which gives an insight into model convergence and generalization and the effectiveness of the regularization strategies.

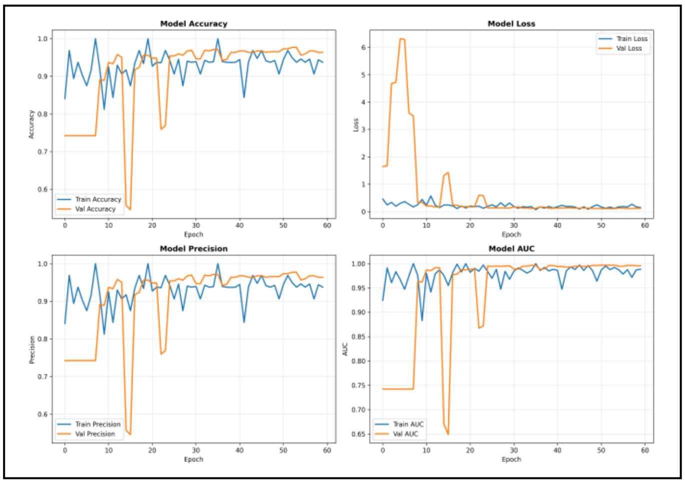

Fig. 3. Training history for MobileNet- CNN with (top-left): accuracy, (top-right): loss, (bottom-left): precision and (bottom-right): AUC for 60 epochs.

The model is able to obtain faster initial convergence in first 10 epochs with the training accuracy greater than 90% and validation accuracy of about 97%. Notably, validation accuracy (orange) sometimes exceeds training accuracy (blue), an odd occurrence that could result from the fact that dropout regularization is enabled during training, but disabled during validation [22].

The loss curves display stable convergence with the validation loss decreasing from around 6.5 to below 0.5 by epoch 15 and then stabilizing. The spike in validation loss at around epoch 10 and its subsequent recovery are evidence of the great success of the ReduceLROnPlateau callback in helping to adjust the learning rates so as to traverse difficult parts of the loss landscape.

Both precision and AUC metrics have a similar trend of convergence with values reaching above 95% by epoch 15 and stability from then on. The close relationship between the training and validation curves among all metrics is indicative of good regularization achieved through dropout (0.2-0.4), L2 weight penalty (λ= 0.001), batch normalization and data augmentation, and hence successful prevention of over-fitting in spite of having 99,842 trainable parameters in the model [23].

Although we set the number of epochs to 60, based on the training curves, we can see that the model was near to optimal performance around epoch 20-25, and the performance at later epochs was nearly flat. The EarlyStopping callback (patience=15) helped make the model explore the loss landscape without excessive computation.

The final test performance (83.97% accuracy, 0.9636 AUC) is very close to the validation performance (97.71% accuracy, 0.9958 AUC in training), however, the difference implies possible validation-set over-fitting or domain shift between the validation and testing data distributions.

6.5. Model Complexity and Scalability

Table 3 compares the architecture complexity and computational requirements of both models, which points out some considerations for implementation.

Table 3. Model Complexity Comparison

Model	Parameters	Training Time (s)	Inference Time (per image)	Memory Footprint
Gaussian Naive Bayes	~20,000	4.73	<0.001 s	~160 KB

Model	Parameters	Training Time (s)	Inference Time (per image)	Memory Footprint
MobileNet-CNN	99,842	247.18	~0.003 s	~390 KB

The Gaussian Naive Bayes classifier with about 20,000 parameters (obtained by 100 principal component analysis components multiplied by two classes and corresponding mean and variance estimations) exhibits remarkable computational efficiency and is therefore applicable in resource-constrained environments. Its inference time of under a millisecond, and memory footprint under 160KB makes it possible to deploy it on edge devices and mobile platforms without any specialized hardware [24].

The MobileNet-CNN architecture, although classified as a "lightweight" architecture and representing only 99,842 parameters, substantially less than the millions of parameters which are a common feature of convolution neural networks used in medical imaging, requires approximately 52 times more training time and three times more inference latency than the Gaussian Naive Bayes model. Nonetheless, its memory footprint of around 390 KB that remains pragmatic for its deployment on contemporary mobile devices, and its latency of around 3ms per image for inference maintains real-time functionality within the clinical workflows [25].

7. Discussion

This section is the interpretation of the findings from the experiment, answers the research questions, and puts the results into context from the general landscape of pneumonia detection.

7.1. Addressing Research Questions

RQ1: Comparative Performance of Gaussian Naive Bayes vs MobileNet-style CNN

The CNN was better than the Gaussian Naive Bayes for all the indicators, with 83.97%accuracy compared with 80.29% and ROC-AUC of 0.9636 compared with 0.8838. Most critically, the recall rate was 98.72, as opposed to 89.23 for Naive Bayes. 98.72 greater than 89.23, this meant far more pneumonia cases were not missed by the CNN. This sensitivity superiority is of prime importance in medical screening with severe clinical implications of false negative results [26]. However, the aggressive positive prediction by CNN led to slightly less precision (80.21% vs 81.12%), producing more false alarming, a clinically acceptable trade off considering false positives can be corrected by follow up examination while missed diagnoses can lead to complications or mortality [2].

RQ2: Trade-offs in Model Selection

Model selection is faced with competing considerations other than predictive accuracy. Naive Bayes provides interpretability by providing explicit class priors and feature likelihoods that can be used to directly inspect probability estimates [27]. The CNN is a black box with 99,842 parameters; the method of explainability, Grad-CAM, could be used to retroactively visualize decision regions [28]. Naive Bayes took 4.73 seconds to complete the training process with a minimum memory requirement (~160 KB), which is suitable for edge devices and resource constraint environments [12]. The CNN in this case took 247 seconds of training time and ~390 KB of memory, which is still practical for modern deployment, but requires 52 times more computational resources [24]. Explicit balanced class weights were taken into consideration in the CNN in the loss function that increases the penalties for minority class misclassified [29]. Naive Bayes has implicitly considered imbalanced data distribution by empirical class priors, though it is limited fundamentally by pixel independence assumption which cannot capture spatial correlations of Pneumonia manifestations [30].

RQ3: Preprocessing Impact on Performance
The CNN maintained native 28×28×1 spatial organization, which allowed convolutional filters to learn translation equivariant features. Naive Bayes worked on flattened PCA-reduced vectors of 784 to 100, and sacrificed the spatial relationships for ease of computation. This preprocessing differential partially accounts for the performance gap because spatial awareness plays an important role in identifying texture patterns and anatomical structures that are indicative of pneumonia [31], [32].

7.2. Clinical Implications and Deployment Considerations

The 98.72% recall of the CNN is a huge clinical benefit for pneumonia screening, especially in resource limited settings with limited radiologist expertise. A two-stage workflow (automated CNN triage and radiologist confirmation) has the potential of reducing workflow and achieving high sensitivity [33].

However, the generalization gap between validation (97.71% accuracy) and test performance (83.97% accuracy) of 14 percentage points raises concerns based on real world applicability. This gap can be because of overfitting to the validation characteristics, distribution shift between splits, or statistical variability given the small test set (624 samples) [6], [34]. External validation using independent data from other institutions and patient demographics is the key for clinical deployment [35].

The 28×28 resolution constraint eliminates fine grained texture information that are critical assets for radiologists to make diagnosis. Clinical grade radiographs are often well over 2000×2000 pixels, and this downsampling likely puts a ceiling on the performance of both models [5], [36].

7.3. Comparison with State-of-the-Art

Our CNN's 83.97% accuracy is below state-of-the-art results on the original Guangzhou dataset: InceptionV3 gave 92.8% [2] and variants of MobileNetV2, 97.4% to 98.2% [9], [37]. This gap is based on three reasons, namely: (1) resolution degradation (28×28 vs 1000×1000 original), (2) limited training data (4,708 vs. datasets with tens of thousands of images), and (3) shallow architecture (99,842 parameters vs. millions in ResNet-50 or DenseNet-121) [38], [39].

In spite of these limitations, our lightweight CNN proves the feasibility of resource-constrained screening applications, and the Naive Bayes baseline confirms that we can still get meaningful detection feasibility from (even simple) statistical models with less sensitivity.

7.4. Generalization Analysis and Overfitting

The CNN's dynamics in training was shown to converge by epoch 20-25 with a tight coupling between training (97%) and validation (98%) accuracy, which displays an effective regularization performed with dropout, L2 penalty, batch normalization, and data augmentation [22]. However, the validation-test performance gap indicates that the model has learned validation specific patterns or distribution shift [6].

Potential explanations include: (1) subtle distributional differences in imaging characteristics between splits despite stratified sampling, (2) indirect validation - caused by hyperparameter tuning set overfitting, (3) statistical variability in small sample sizes, or (4) domain shift induced by training only data augmentation [34], [40]. The solution to this would likely be some external validation, more training data, ensemble methods, or domain adaptation techniques [41].

7.5. Limitations

PneumoniaMNIST is a single pediatric dataset from a single institution. The presentation of pneumonia differs between age groups and between etiological causes and populations, which may hamper the transferability of models [42] . The binary task (normal vs pneumonia) is an oversimplification of clinical reality where radiologists differentiate between bacterial/viral pneumonia and other types of pulmonary pathologies [43]. Resolution Constraints Due to the 28×28 limitation, the fidelity to diagnose is fundamentally limited as compared to clinical grade full-resolution radiographs [44]. Neither model gives spatial localization of findings. Clinical deployment requires visual explanations of pathological areas to allow verification by a radiologist [45]. Chest radiograph interpretation is subject to inter-observer variability (Cohen's kappa 0.4-0.6), which implies that training labels are noisy and that better performance is not possible [46].

8. Conclusion

The present study performs a systematic comparative evaluation of a gaussian naive bayes classifier and a lightweight mobile net inspired convolutional neural network for pneumonia detection for the PneumoniaMNIST dataset. The idea is to find out if deep learning strategies provide performance advantages that are large enough compared to statistically transparent baselines to justify their computational costs in medical imaging resource-constrained environments.

Empirical results show that the convolutional architecture significantly outperforms the Naive Bayes model in terms of all the evaluated metrics and it achieves higher values of accuracy, precision, recall, F1-score and area under the receiver-operator characteristic curve on the held-out test partition. In particular, the network's higher recall means fewer false negatives; an important factor in clinical screening where false negatives have severe health consequences. This superiority is credited to the network's ability to preserve and utilize spatial image structure through convolution based feature extraction, where the Naive Bayes classifier which works on the flattened dimension reduced feature vectors, rejects spatial relationships which are crucial for accurate pneumonia inference.

As a result, however, such performance improvements come with salient trade-offs. The convolutional model requires much longer training times and memory footprint than the baselines given by the Naive Bayes algorithm, and is an opaque black-box system intrinsically impeding clinical interpretability. Moreover, the recorded generalization difference between the validation and test data sets, as well as the sub-optimal performance compared to current state-of-the-art approaches on full-resolution data sets, point to constraints imposed by the low resolution, low sample size and simplification to binary classification of the data sets.

Future studies should focus on the external validation with independent cohorts that are sampled from a range of institutions and patient demographics to assess the ability for generalisation in a robust manner. Integration of transfer learning, ensemble modelling and explainability methods like Grad-CAM. Weighted Class Activation Mapping would overcome existing shortcomings by augmenting feature representations and providing spatial saliency instant visualization for clinical corroboration. Multi-task learning architectures and uncertainty quantification techniques are an additional potential field for simultaneously improving predictive performance and ensuring clinical reliability.

In summary, lightweight deep learning architectures provide the tangible performance benefit compared to classical statistical models for pneumonia detection and even within the constraints of limited computational resources and limited data availability. The choice between an interpretable baseline and a high-performance neural network should therefore be guided by the application-specific priorities in question, e.g. required sensitivity of diagnostics, available computational capacity, interpretability requirements and deployment constraints. For medical - imaging tasks where the level of diagnostic accuracy directly impacts patient outcomes, the added discriminative power of the CNN provides sufficient justification for the added computational burden, provided the external validation and explainability mechanisms are robust to ensure generalization and secure clinician trust. Broadly, these results suggest that computationally efficient solutions of deep learning algorithms can be used to detect pneumonia in clinically relevant settings on low resolution images, supporting their feasibility for point-of-care diagnostic systems and resource-limited healthcare settings where automated screening could help to increase access to diagnostic services.

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